

DO's & DON'TS FOR THE LATHE

DO's

- ✓ Do Become Acquainted with the function of the important parts of lathe
- ✓ Do keep your machine properly lubricated
- ✓ Do keep machines clean and orderly. A dirty machine is not conducive to good workmanship.
- ✓ Do thoroughly understand & plan the job before starting to work on machine.
- ✓ Do keep the cutting tool sharp, dull cutting tools require a longer time to do the same job and give a poor finish and put the machine under an unwarranted strain.
- ✓ Do take as heavy a cut as a machine, work and cutting tool will permit. A series of light cuts wastes time and make necessary work for the operator.
- ✓ Do take interest in your work with responsibility.

DON'TS

- * Don't wear loose fitting shop coats or aprons when operating any machine.
- * Don't ever leave the chuck wrench in the chuck.
- * Don't push any lever or turn any handle on a lathe unless you know what will happen as a result.
- * Don't wear long ties, finger rings while operating machine.
- * Don't try to run a machine & engage in conversation at the same time. If you must talk then shut down the machine.
- * Don't be afraid to wear goggles when turning work which produces flying chips.
- * Don't attempt to check the hole when boring, without first covering the boring tool to guard against arm and hand injuries.
- * Don't put your hand or fingers on any revolving work or toll at any time.
- * Don't go away and leave your machine running. If you must leave then shut down.
- * Don't drop chucks, face or drive plates or lay work or tools on the ways of the lathe
- * Don't offer excuses when you scrap a job. Accept your responsibility and try to do better the next time.

LATHE

1.1 Introduction

A lathe is a machine tool which spins a block of material to perform various operations such as cutting, sanding, knurling, drilling, etc. or deformation with tools that comes in contact with the work piece to create an object which has symmetry about an axis of rotation. Lathes are used in woodturning, metalworking, metal spinning, and in glass working. Lathes can be used to shape pottery as well. Most suitably metalworking lathes can be used to produce most solids of revolution, plane surfaces and screw threads or helices. The material can be held in place by either one or two centers, at least one of which can be moved horizontally to accommodate varying material lengths. Ornamental lathes can produce three-dimensional solids of incredible complexity. Machine shop personnel must be thoroughly familiar with the lathe and its operations to perform various tasks.

1.2 Principal of Operations

The lathe is a machine tool used principally for shaping articles of metal (and sometimes wood or other materials) by causing the work piece to be held and rotated by the lathe while a tool bit is advanced into the work causing the cutting action. The basic lathe that was designed to cut cylindrical metal stock has been developed further to produce screw threads, tapered work, drilled holes, knurled surfaces, etc. The typical lathe provides a variety of rotating speeds and a means to manually and automatically move the cutting tool into the work piece.

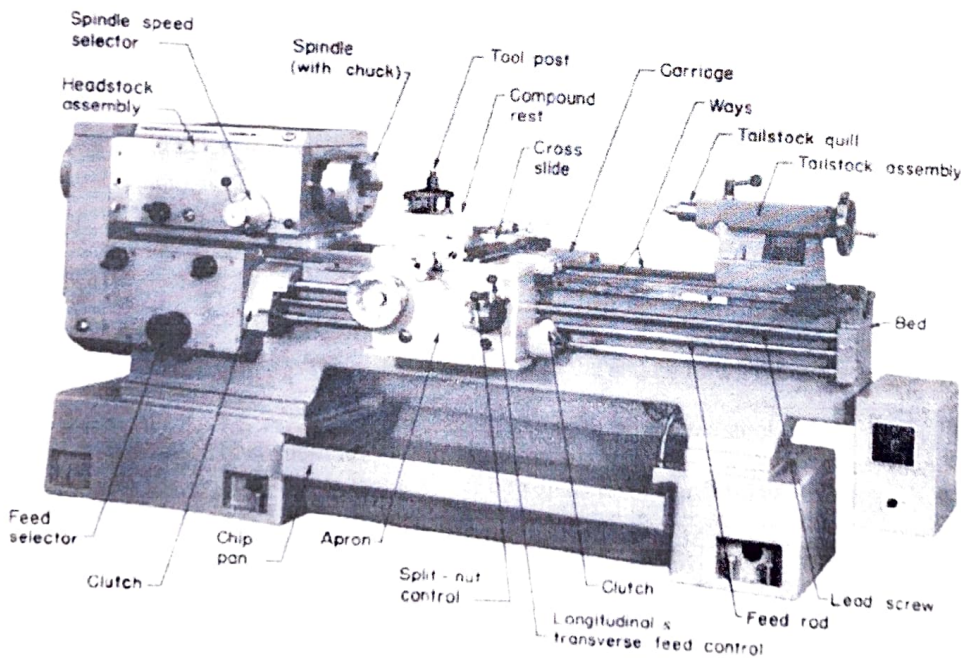
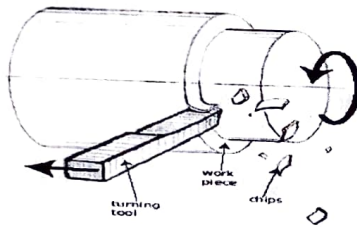


Fig.1 Engine Lathe

1.3 Lathe and its Parts

Headstock	Contains the spindle in two preloaded ball bearings
Spindle	The spindle is inside the headstock and is driven with a belt running from the motor pulley to a pulley on the rear end of the spindle shaft. The nose of the spindle is treaded on the outside to receive chucks and tapered on the inside to receive other accessories.
Chucks	A 3-jaw or 4-jaw chuck threads onto the spindle nose to hold your work, a drill chuck are used on the tailstock to center drill your part.
Tool Post	Attaches to the lathe table and holds a 1/4" square cutting tool
Cross Slide Table	Also sometimes spelled "cross slide," it is the table with two T-slots that holds the tool post.
Cross Slide Gib	A tapered plastic wedge that is held in place by a gib lock. It fits between the angled surfaces of the dovetail and is used to adjust for wear. As wear occurs and the table develops "slop," the lock is loosened and the gib is pushed further into the gap, taking up any play. This allows the machine to always be kept in peak adjustment.
Tail Stock spindle	Has a #0 Morse internal taper for holding chucks and other tools. A hand wheel moves it in and out for drilling
Tail Stock Locking Screw	Locks the tailstock in place on the bed to keep it from moving. When loosened, the tailstock can be slid up and down the bed.
Bed	The dovetailed steel bar that the saddle and tailstock are moved back and forth on.
Saddle	The part that supports the cross slide table and is moved up and down the bed using the lead screw hand wheel.
Saddle Gib	Functions like the cross slide gib to keep the saddle in tight adjustment against the dovetailed bed.
Lead Screw	The threaded screw under the bed that controls movement of the saddle. A "saddle nut" underneath attaches the bed to the leadscrew. Turning the lead screw hand wheel moves the saddle down the bed.
Tail Stock Gib	A brass part attached to the base of the tailstock that runs on one of the bed dovetails. The brass part is expected to wear rather than the more expensive bed and can be adjusted for tightness as it wears.
Lathe Base	The cast metal base upon which the lathe bed and headstock sit.
Drawbolt	Goes through the hole in the spindle to draw chucks and other accessories into the headstock taper inside the spindle. A special washer locates it on center in the spindle hole.

#1 Morse Arbor	The tailstock drill chuck normally has a #0 Morse arbor threaded into the back of it for use in the tailstock spindle. That arbor can be removed and replaced with the #1 Morse arbor so the drill chuck can be used in the headstock.
Dead Centers	#1 and #0 Morse arbors have a 60° point and are used to locate and hold work "between centers" on the lathe. The #1 Morse arbor rotates with the headstock, but because the tailstock spindle does not rotate, the rear #0 Morse arbor is called a "dead" center. This needs to be kept lubricated because it creates friction with the moving part it is locating. Most machinists eventually replace this with a "live" center that turns on a ball bearing.
Tommy Bars	Round steel bars used to tighten and loosen chucks and other spindle accessories. Sometimes called "Spindle Bars."
Faceplate	A cast plate that threads onto the spindle nose. A work piece can be bolted to it as an alternative to using a chuck. It has three slots to drive a drive dog.
Drive Dog	Also called a "Lathe Dog," this part is attached to a piece of bar stock by means of a screw that goes through the side and the long point is placed into one of the slots in the faceplate. The part is located between the lathe centers (live or dead) and when the faceplate turns, the dog actually drives the piece to rotate it for cutting. It also acts as a universal joint when turning a part between centers when the headstock is rotated to a slight angle, allowing a tapered part to be cut.
Head Stock Locking Screw	Holds the headstock in place. The screw is a pointed set screw. The point engages a tapered groove in the pin that sticks up out of the lathe bed. When the screw is tightened, it pulls the headstock down onto the alignment key and holds it tight against the lathe bed.
Alignment Key	A precision ground key that fits in slots in the top of the bed and bottom of the headstock to keep the headstock aligned straight with the tailstock. Removing this key and rotating the headstock allows tapers to be cut.
V-belt	A Kevlar-reinforced Urethane belt that drives the spindle by means of the pulleys.
2-Position Pulley	The motor turns a maximum of about 6000 RPM. Putting the drive belt in the normal (rear) position gears the motor down about 2:1 for a maximum speed of about 2800 RPM. The "High Torque" position (closest to the headstock) gears it about 4:1 for lower speed but more torque when needed for heavy cuts.
Variable Speed Control Knob	Controls motor speed from 0 to 2800 RPM

1.4 Lathe Dimensions

Lathe size is determined by the swing and length of bed. Swing indicates the largest diameter that can be turned over the ways (flat or v-shaped bearing surface that aligns and guides moveable parts of machines). Bed length is entire length of the ways. Bed length must not be mistaken for the maximum length of the work that can be turned between centers.

The longest piece that can be turned is equal to the length of the bed minus the distance taken up by the headstock and tailstock.

When comparing the size and working capacities of metal lathes there are several key dimensions to consider:

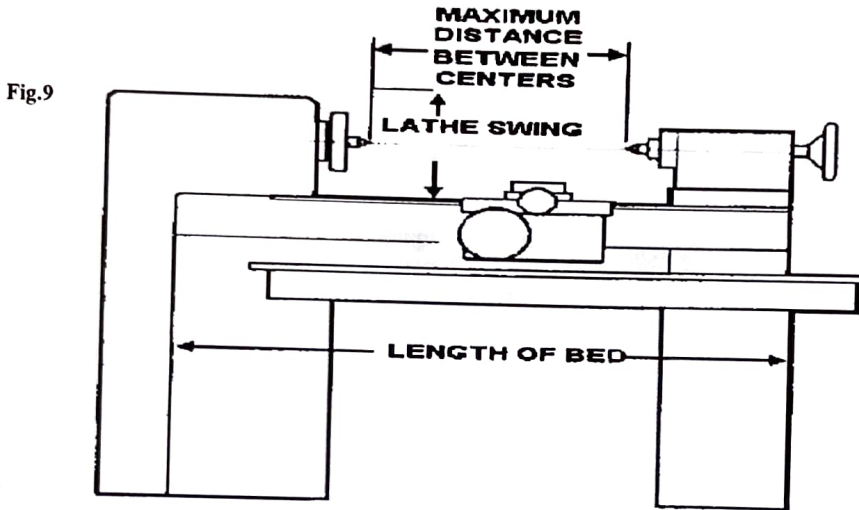


Fig 2: Dimensions of Lathe

Swing over bed:

The diameter of the largest workpiece that can be rotated on the spindle without hitting the bed. This is the first of the two numbers used to describe the size of a metal lathe.

Distance between Centers:

The longest piece of work that can be held between a center in the headstock and a center in the tailstock. (See glossary below for more information). This is the second of the two numbers used to describe the lathe size.

Swing over the Carriage:

The diameter of the largest workpiece that can rotate over the carriage without hitting it. On the 9x lathes this is about 5"

Diameter of Spindle through hole:

The diameter of the hole that passes through the spindle. On the 9x lathes (or any lathe having a #3 Morse Taper spindle) it is about 3/4". When facing relatively long stock, the free end of the stock can pass through the spindle if it is no larger than the through-hole diameter.

1.5 Types of Lathes:

1. Engine Lathe

The most common form of lathe, motor driven and comes in large variety of sizes and shapes.

2. Bench Lathe

A bench top model usually of low power used to make precision machine small work pieces.

3. Tracer Lathe

A lathe that has the ability to follow a template to copy a shape or contour.

4. Automatic Lathe

A lathe in which the work piece is automatically fed and removed without use of an operator. Cutting operations are automatically controlled by a sequencer of some form.

5. Turret Lathe

Lathes which have multiple tools mounted on turret either attached to the tailstock or the cross-slide, which allows for quick changes in tooling and cutting operations.

6. Computer Controlled Lathe

Highly automated lathes, where cutting, loading, tool changing, and part unloading are automatically controlled by computer coding.

1.6 Machining Parameters:

Cutting Speed:

The speed in surface feet per minute or meters per minute at which the metal may be machined efficiently

Lathe Cutting Speed:

It may be defined as the rate at which a point on the work circumferences travels past the cutting tool. When work is machined in a lathe, it must be turned at a specific number of revolutions per minute (r/min), depending on its diameter, to achieve the proper cutting speed.

Cutting speed is always expressed in feet per minute (ft/min) or in meters per minute (m/min).

Lathe Feed:

The feed of a lathe may be defined as the distance the cutting tool advances along the length of work for every revolution of spindle.

Feed of the lathe is dependent on the speed of the lead screw or feed rod. Speed is controlled by the change gears in the quick – change gear box.

Cutting Speed:

Cutting speed is how fast the metal comes into contact with the tool at the cutting point. On a lathe, it is the rate at which the surface of the job passes the cutting tool. This takes into account the diameter of the job. The general formula for a cutting speed is as follows:

Feed:

The feed of a lathe may be defined as the distance the cutting tool advances along the length of the work for every revolution of the spindle.

1.7 Various operations performed on Lathe machine

Turning	:	Produces straight, conical, curved, or grooved work pieces
Facing	:	Produces a flat surface at the end of the part
Boring	:	To enlarge a hole
Drilling	:	To produce a hole
Cutting off	:	To cut off a work piece
Threading	:	To produce threads
Knurling	:	Produces a regularly shaped roughness
Profiling	:	To turn cylindrical work pieces with rough and finished cuts
Grooving	:	To make furrows or channels

1.8 Operations on a Lathe:

Turning:

cutting tool is moved parallel to the axis of the work piece to produce a cylindrical surface by removing the unwanted material in the form of chips. Here depth of the cut is given by moving the tool perpendicular to the lathe axis.

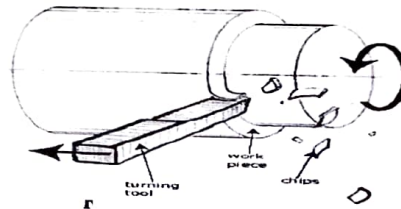


Fig.3 Plain Turning

Facing:

It is an operation to produce flat surface on the ends of the work piece. Here the cutting tool is fed against the rotating work piece perpendicular to the lathe axis and the depth of the cut is given by moving the tool parallel to the lathe axis.

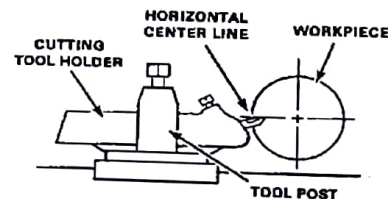


Fig.4 Facing operation

Taper Turning :

It is an operation to produce conical surface on the work piece.
Methods for taper turning.

- ✓ By swivelling the compound rest
- ✓ By offsetting the tail stock
- ✓ By taper turning attachment
- ✓ By using form tool

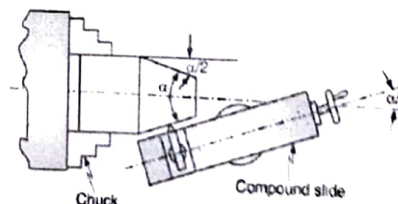


Fig.5 Swivelling of Compound Rest

